



**TECHNICAL UNIVERSITY
OF CLUJ-NAPOCA**

Faculty of Electronics, Telecommunications and Information Technology

CAD Techniques

-Project-

**CIRCUIT FOR CONTROLLING THE CONCENTRATION OF
CARBON MONOXIDE IN AN ENCLOSURE**

Coordinating teacher:

Dr. Eng. Raul Traian-Fizeșan

Student:

Vărgău Paul-Inocențiu

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-INTRODUCTION-

1. Requirements

Task: Design a system using resistive gas sensors to maintain the concentration of carbon monoxide in an enclosure between the limits specified. There is a source in the enclosure that continuously generates carbon monoxide. When the concentration has reached the upper limit, the system will start the fan that will introduce fresh air. When the concentration of carbon monoxide reaches the lower limit the system will give the command to stop the fan. From the sensor data sheet it is known that at a variation of the gas concentration specified. The electrical resistance of the sensor varies linearly in the range specified. The variation of the electrical resistance of the sensor must be converted into a voltage variation in the range $[2 \div (V_{cc}-2V)]$. The fan is controlled by a hysteresis comparator via a relay which is modeled with a resistor. The state of the fan (on/off) is indicated by an LED of the color specified.

Table 1 – The parameters of the circuit

Indoor carbon monoxide concentration [ppm]	The measurement range of the sensor [ppm]	Sensor resistance [k Ω]	V _{cc} [V]	LED color
100-10000	70-16000	80-10	16	Red

2. Theoretical Support

This circuit is designed to monitor and control the concentration of carbon monoxide (CO) within an enclosed space. The system uses a carbon monoxide sensor to detect CO levels and activates a ventilation fan to reduce the CO concentration when it exceeds a predefined threshold. An alert system is also incorporated to warn occupants of hazardous CO levels.

The components needed for this circuit and their brief description are listed down below:

1. Resistor-Based Carbon Monoxide Sensor (R_s): calculates the carbon monoxide concentration in the enclosure.
2. Operational Amplifiers (Op-Amps): used to build circuit blocks consisting of various functions.
3. Relay module: used as a switch for the ventilation system.
4. Alert system: A Light-Emitting-Diode (LED) used to warn the user that the CO level in the enclosure is too high and that the ventilation system was turned on.
5. Power Supply: Needed to provide power to the Op-Amps and the other components of the circuit.

The next part of this paper will focus on the basic theoretical fundamentals needed for the circuit design.

2.1 The Voltage Divider

A voltage divider (Figure 1 – Voltage Divider Circuit

Figure 1 – Voltage Divider Circuit) is a simple and commonly used electronic circuit which consists in a combination of two resistors supplied by a voltage source to create an output voltage which is lower than the input voltage. The output voltage is often called Reference Voltage (V_{REF}). The formula for this circuit is stated in Eq. 1.

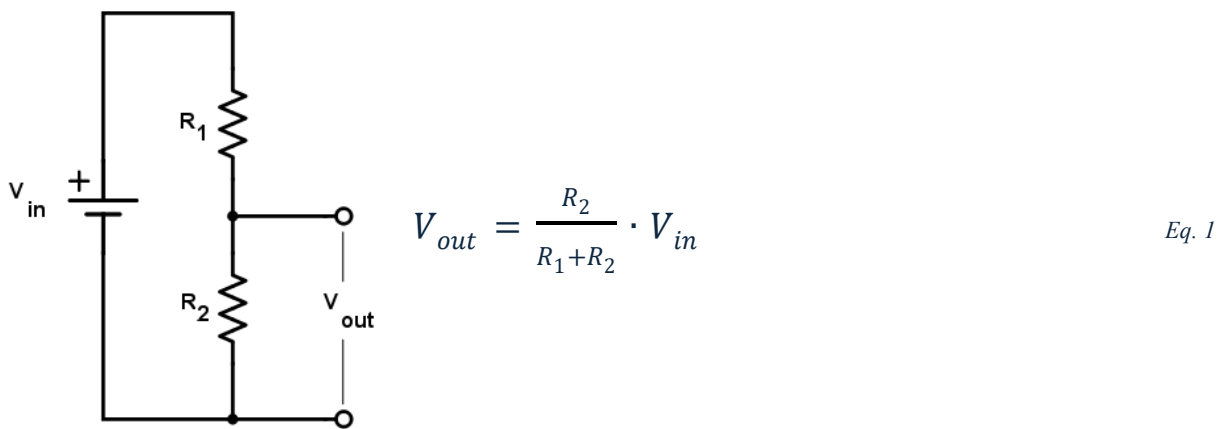
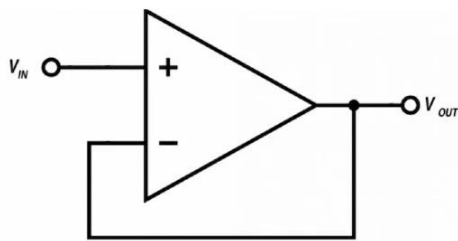


Figure 1 – Voltage Divider Circuit

Figure 2 – Voltage Divider Circuit

2.2 The Voltage Follower

A voltage follower, also known as a buffer amplifier, is an electronic circuit that provides unity gain (a gain of 1) while offering high input impedance and low output impedance. The formula for this circuit is stated in Eq. 2. [1]



$$V_{out} = V_{in} \quad \text{Eq. 2}$$

Figure 3 - Voltage Follower

2.3 The Domain Converter

A domain converter (Figure 4 – Inverting Voltage Domain Converter, refers to a device or circuit that converts signals from one domain to another. The term "domain" can refer to various aspects such as voltage levels, current levels, or different types of signals. [1]

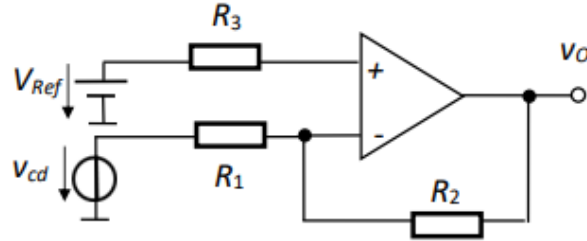


Figure 4 – Inverting Voltage Domain Converter

For the inverting configuration, the minimum/maximum command voltage leads to the maximum/minimum output voltage:

$$V_{cd,min} \rightarrow V_{O,max}; V_{cd,max} \rightarrow V_{O,min}$$

Eq. 3

$$V_{O,max} = \left(1 + \frac{R_2}{R_1}\right) \cdot V_{Ref} - \frac{R_2}{R_1} \cdot V_{cd,min}$$

Eq. 4

$$V_{O,min} = \left(1 + \frac{R_2}{R_1}\right) \cdot V_{Ref} - \frac{R_2}{R_1} \cdot V_{cd,max}$$

Eq. 5

$$\frac{R_2}{R_1} = \frac{V_{O,max} - V_{O,min}}{V_{cd,max} - V_{cd,min}}$$

Eq. 6

2.4 The Hysteresis Comparator

A hysteresis comparator (Figure 5 - Non-Inverting Asymmetric Hysteresis Comparator) is a signal processing system that is often used on noisy signals, the output switching between V_{OL} and V_{OH} everytime it meets the threshold voltages. [1]

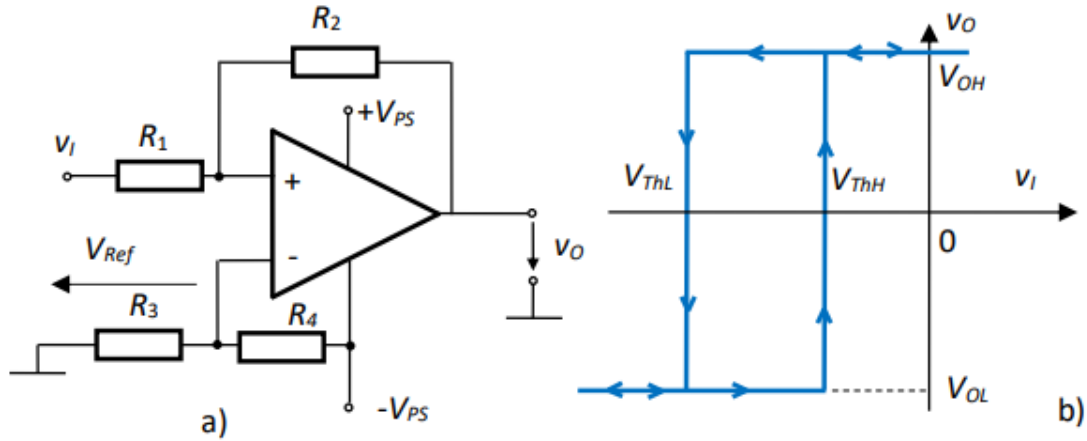


Figure 5 - Non-Inverting Asymmetric Hysteresis Comparator

The formulas for V_{ThH} and V_{ThL} are stated below. (Eq. 7, Eq. 8)

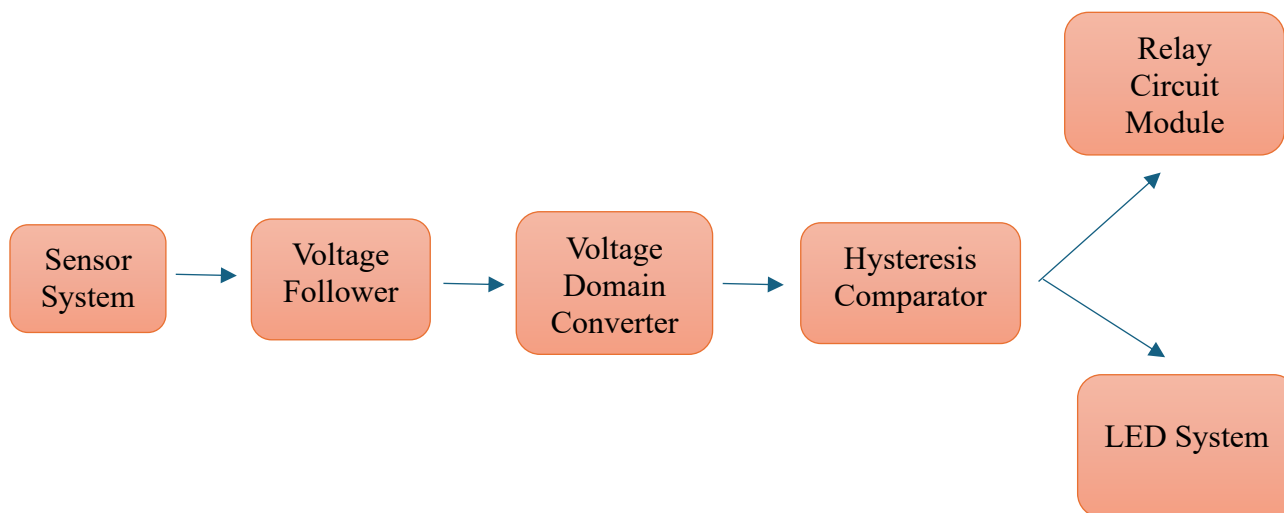
$$V_{ThH} = \left(1 + \frac{R_1}{R_2}\right) \cdot V_{Ref} - \frac{R_1}{R_2} \cdot V_{OL}$$

Eq. 7

$$V_{ThL} = \left(1 + \frac{R_1}{R_2}\right) \cdot V_{Ref} - \frac{R_1}{R_2} \cdot V_{OH}$$

Eq. 8

BLOCK DIAGRAM AND CIRCUIT FUNCTIONALITY



1. Block Diagram Explanation

Sensor System

- Components: Resistors R1 (sensor) and R2.
- Function: This part of the circuit forms a voltage divider that senses the concentration of carbon monoxide. The resistance $\{r_s\}$ changes based on the carbon monoxide level, resulting in a corresponding voltage change at the input.

Voltage Follower

- Components: Operational amplifier U1, resistors R3, R4, R5.
- Function: Buffers the input signal from the sensor system to prevent loading and to provide a stable voltage (V_{cd}) for the next stage. It ensures the input impedance is high and the output impedance is low.

Voltage Domain Converter

- Components: Operational amplifier U2, resistors R6, R7.
- Function: Converts the buffered signal to a specific voltage domain. In this case, it maps the sensor output to a usable voltage range (2V to 14V).

Hysteresis Comparator

- Components: Operational amplifier U3, resistors R8, R17, R18, R19.
- Function: Implements hysteresis to provide stable switching behavior. It compares the signal from the domain converter and controls the relay circuit based on preset thresholds to avoid noise-induced false triggers.

Relay Circuit Module

- Components: Transistor Q1, resistors R14, R15, R16.
- Function: Uses the output from the hysteresis comparator to drive the relay. When the comparator output (compout) activates the transistor (Q1), it powers the relay, which controls the ventilation system to clean the air by activating the ventilation system.

LED System

- Components: Resistor R13, LED D1.
- Function: The LED system is also controlled by the hysteresis comparator output. It provides a visual indication of the carbon monoxide levels. When the comparator output indicates a high level of carbon monoxide, the LED system is activated to alert the user that the ventilation system is on.

2. Circuit Blocks and Calculus

First of all, we need to calculate the voltage thresholds where the circuit switches. For that, we use the mathematical formulas for the slope calculation, taking into account the requirements of the circuit specified at page 3, in the domain given to us.

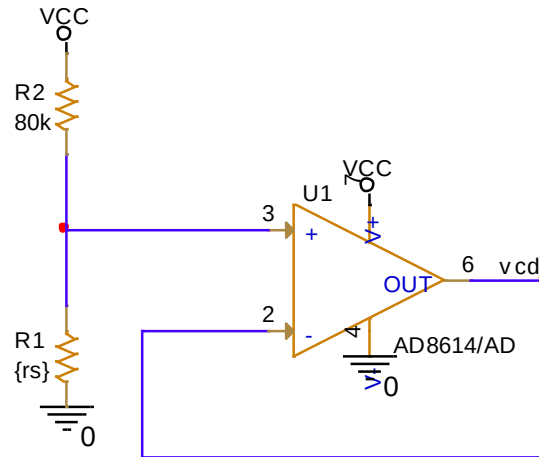
$$V_{ThH} = (V_{CC} - 2) \cdot \frac{[(V_{CC} - 2) - 2] \cdot (Concentration_{min} - SensorRange_{min})}{SensorRange_{max} - SensorRange_{min}} \Rightarrow$$

$$\Rightarrow V_{ThH} = 14 - \frac{12 \cdot 30}{15930} \Rightarrow V_{ThH} = 13.97V$$

$$V_{ThL} = (V_{CC} - 2) \cdot \frac{[(V_{CC} - 2) - 2] \cdot (Concentration_{max} - SensorRange_{min})}{SensorRange_{max} - SensorRange_{min}} \Rightarrow$$

$$\Rightarrow V_{ThH} = 14 - \frac{12 \cdot 9930}{15930} \Rightarrow V_{ThH} = 6.52V$$

To simulate the sensor and power the domain converter with a certain stable command voltage (V_{cd}), a voltage divider to create a reference voltage which equals the voltage at the output of the buffer was used, and a voltage follower to ensure that the output impedance is low.



PARAMETERS:

$r_s = 10k$

Figure 6 - Buffer with V_{ref}

$$v^+ = v^- \Rightarrow$$

$$\Rightarrow V_{Ref1} = V_{cd} \Rightarrow$$

$$\Rightarrow \frac{R_1}{R_1 + R_2} \cdot V_{CC} = V_{cd} \Rightarrow$$

$$\Rightarrow \begin{cases} V_{cd_min} = \frac{10k}{10k + 80k} \cdot 16 \Rightarrow V_{cd_min} = 1.78V \\ V_{cd_max} = \frac{80k}{80k + 80k} \cdot 16 \Rightarrow V_{cd_max} = 8V \end{cases}$$

The domain converter takes the output voltage of the buffer at the inverting input and a reference voltage at the non-inverting input, and the resistors are chosen such that, through the negative feedback, the output voltage of the converter (V_{dcout}) takes values in the [2V;14V] interval.

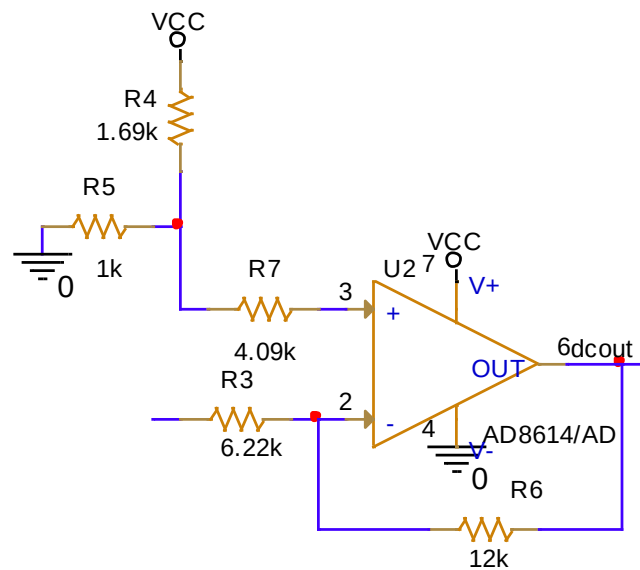


Figure 7 - Domain Converter

$$\frac{R_6}{R_3} = \frac{V_{O_max} - V_{O_min}}{V_{cd_max} - V_{cd_min}} \Rightarrow$$

$$\Rightarrow \frac{R_6}{R_3} = \frac{14 - 2}{8 - 1.78} \Rightarrow$$

$$\Rightarrow \frac{R_6}{R_3} = \frac{12}{6.22} \Rightarrow$$

$$\Rightarrow \frac{R_6}{R_3} = 1.93$$

The value chosen for R_6 is 12k Ω , and for R_3 is 6.22k Ω .

$$V_{Ref2} = \frac{V_{O_min} + \frac{R_6}{R_3} \cdot V_{cd_max}}{1 + \frac{R_6}{R_3}} \Rightarrow$$

$$\Rightarrow V_{Ref2} = \frac{2 + 1.93 \cdot 8}{1 + 1.93} = 5.95V$$

$$\begin{aligned} V_{Ref2} &= \frac{R_5}{R_4 + R_5} \cdot V_{CC} \Rightarrow \\ \Rightarrow \frac{R_5}{R_4 + R_5} \cdot 16 &= 5.95 \Rightarrow \\ \Rightarrow 10.05 \cdot R_5 &= 5.95 \cdot R_4 \Rightarrow \\ \Rightarrow R_4 &= 1.69 \cdot R_5 \end{aligned}$$

The value chosen for R_4 is $1.69k\Omega$, and for R_5 is $1k\Omega$.

$$\begin{aligned} R_7 &= R_3 || R_6 \Rightarrow \\ \Rightarrow R_7 &= \frac{R_3 \cdot R_6}{R_3 + R_6} \Rightarrow \\ \Rightarrow R_7 &= 4.09k\Omega \end{aligned}$$

The resistor R_7 is used to flatten the undesired noise of the circuit block through impedance matching.

By simulating a DC Sweep on the domain converter's output voltage, having the sensor resistor set as a global parameters which takes values from $10k\Omega$ to $80k\Omega$, as stated in the requirements, and setting cursors at the threshold voltages (V_{ThL} , V_{ThH}), we get the values for the sensor resistances where the circuit switches, which are going to help us later in the calculations.

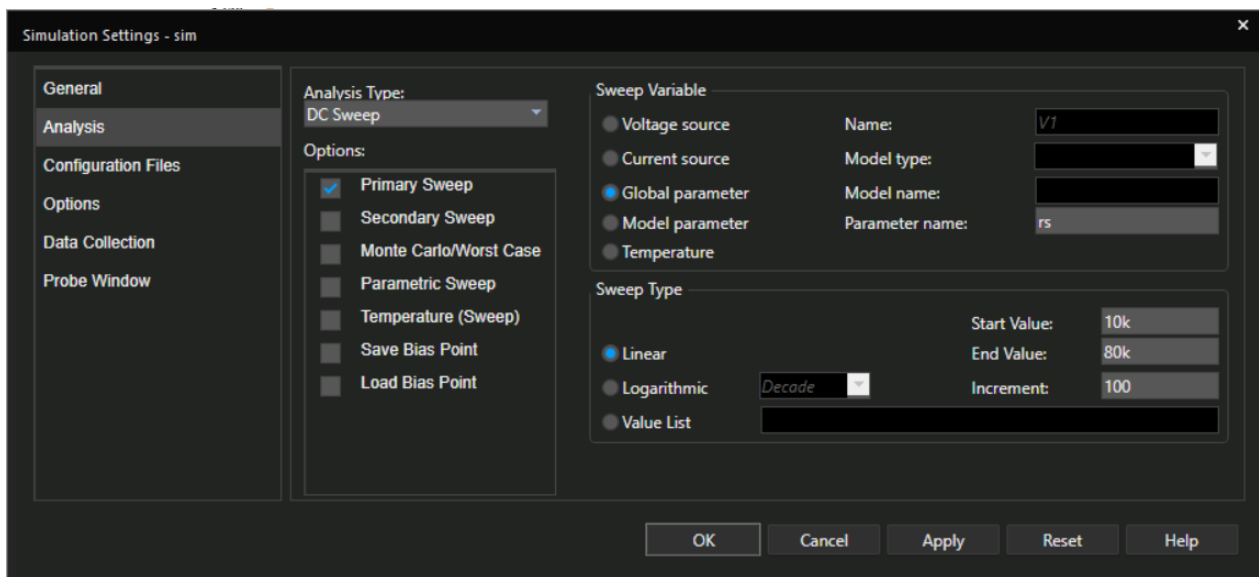


Figure 8 - Simulation Profile

In Figure 9 - Measurements, we can see that at $\sim 13.97\text{V}$, the resistance of the sensor is $\sim 10.1\text{k}\Omega$, and at $\sim 6.52\text{V}$, the resistance of the sensor is $\sim 43.6\text{k}\Omega$.

Trace Color	Trace Name	Y1	Y2	Y1 - Y2
	X Values	10.099K	43.645K	-33.547K
CURSOR 1,2	V(DCOUT)	13.962	6.5262	7.4362

Figure 9 - Measurements

The non-inverting asymmetric hysteresis comparator is essential for providing reliable and stable switching behavior in the carbon monoxide detection circuit. It ensures that the output only changes state when there is a significant change in the input signal, providing noise immunity, debouncing, and accurate signal conditioning.

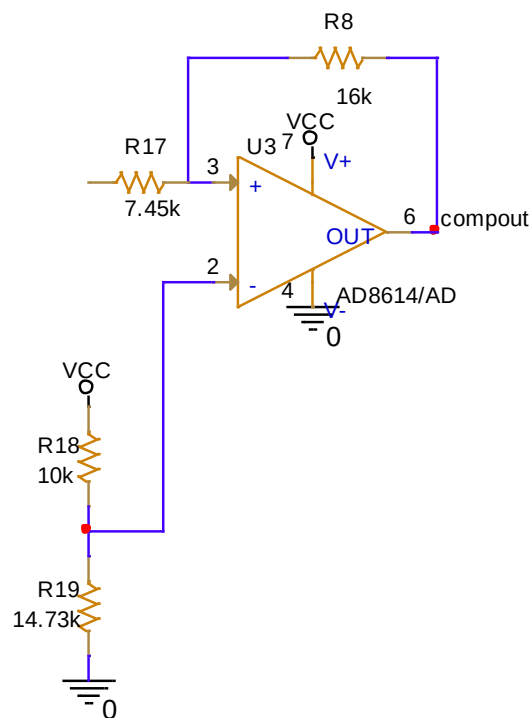


Figure 10 - Non-Inverting Asymmetric Hysteresis Comparator\

$$\begin{cases} V_{ThL} = -\frac{R_1}{R_2} \cdot V_{OH} + \left(1 + \frac{R_1}{R_2}\right) \cdot V_{Ref3} \\ V_{ThH} = -\frac{R_1}{R_2} \cdot V_{OL} + \left(1 + \frac{R_1}{R_2}\right) \cdot V_{Ref3} \\ V_{OL} = 0 \end{cases} \Rightarrow$$

$$\Rightarrow V_{ThL} = -\frac{R_1}{R_2} \cdot V_{OH} + V_{ThH} \Rightarrow$$

$$\Rightarrow 6.52 = -\frac{R_1}{R_2} \cdot 16 + 13.97 \Rightarrow \frac{R_1}{R_2} = \frac{7.45}{16} \Rightarrow \begin{cases} R_1 = 7.45k\Omega \\ R_2 = 16k\Omega \end{cases}$$

The output block circuit consists in a BJT transistor (Q2N2222) commonly used in electronics because of its low-to-medium voltage applicability, price and functionality. In the base of the transistor, R16 is used to limit the current injected to a safe value that will not damage the component. When the hysteresis comparator switches, the transistor reaches its saturation state where it allows the current to flow from the collector to the emitter, so that V_{CC} will be able to power up the LED and, through another voltage divider needed to limit the voltage drop on the relay from 16V to 5V, powering up the relay.

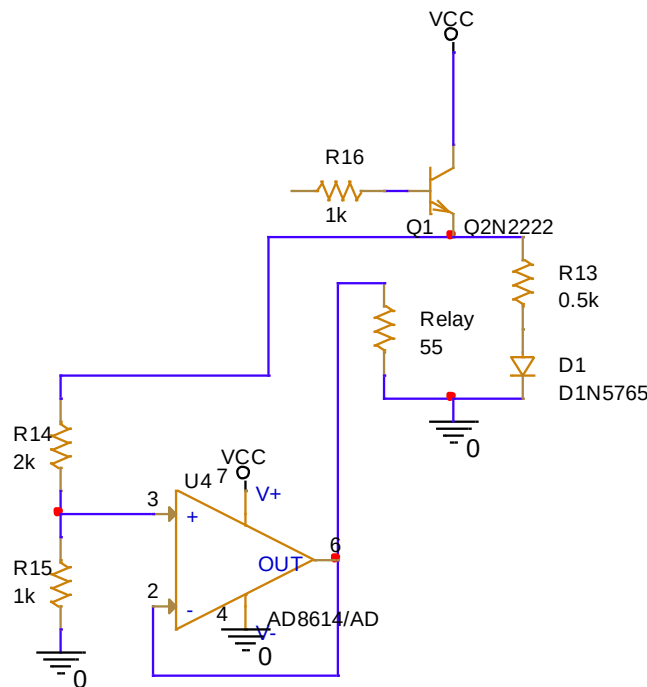


Figure 11 - Relay and LED module

$$\frac{R_{15}}{R_{15} + R_{14}} \cdot (V_{CC} - V_{CEsat}) = V_{Relay} \Rightarrow$$

$$\Rightarrow \frac{1}{2 + 1} \cdot (16 - 1) = V_{Relay} \Rightarrow$$

$$\Rightarrow V_{Relay} = 5V$$

This voltage divider was used to limit the voltage of the relay to a value known, stated in the datasheet. The relay used is a Songle SRD-05VDC-SL-C.

6. COIL DATA CHART (AT20 ° C)

Coil Sensitivity	Coil Voltage Code	Nominal Voltage (VDC)	Nominal Current (mA)	Coil Resistance (Ω) □ 10%	Power Consumption (W)	Pull-In Voltage (VDC)	Drop-Out Voltage (VDC)	Max-Allowable Voltage (VDC)
SRD (High Sensitivity)	03	03	120	25	abt. 0.36W	75%Max.	10% Min.	120%
	05	05	71.4	70				
	06	06	60	100				
	09	09	40	225				
	12	12	30	400				
	24	24	15	1600				
	48	48	7.5	6400				
SRD (Standard)	03	03	150	20	abt. 0.45W	75% Max.	10% Min.	110%
	05	05	89.3	55				
	06	06	75	80				
	09	09	50	180				
	12	12	37.5	320				
	24	24	18.7	1280				
	48	48	10	4500	abt. 0.51W			

Figure 12 - Relay Datasheet Information

The red LED (D1N5765) is a current controlled LED, with it's operating region being between 20mA and 35mA, as stated in the datasheet. A 0.5kΩ resistor was chosen to limit the current flowing to the LED to a value in which it would power up and not be damaged.

$$\begin{aligned}
 I_{LED} &= \frac{V_{CC} - V_{CEsat}}{R_{13}} \Rightarrow \\
 \Rightarrow 30mA &= \frac{16 - 1}{R_{13}} \Rightarrow \\
 \Rightarrow R_{13} &= 0.5k\Omega
 \end{aligned}$$

3. Full Circuit Design

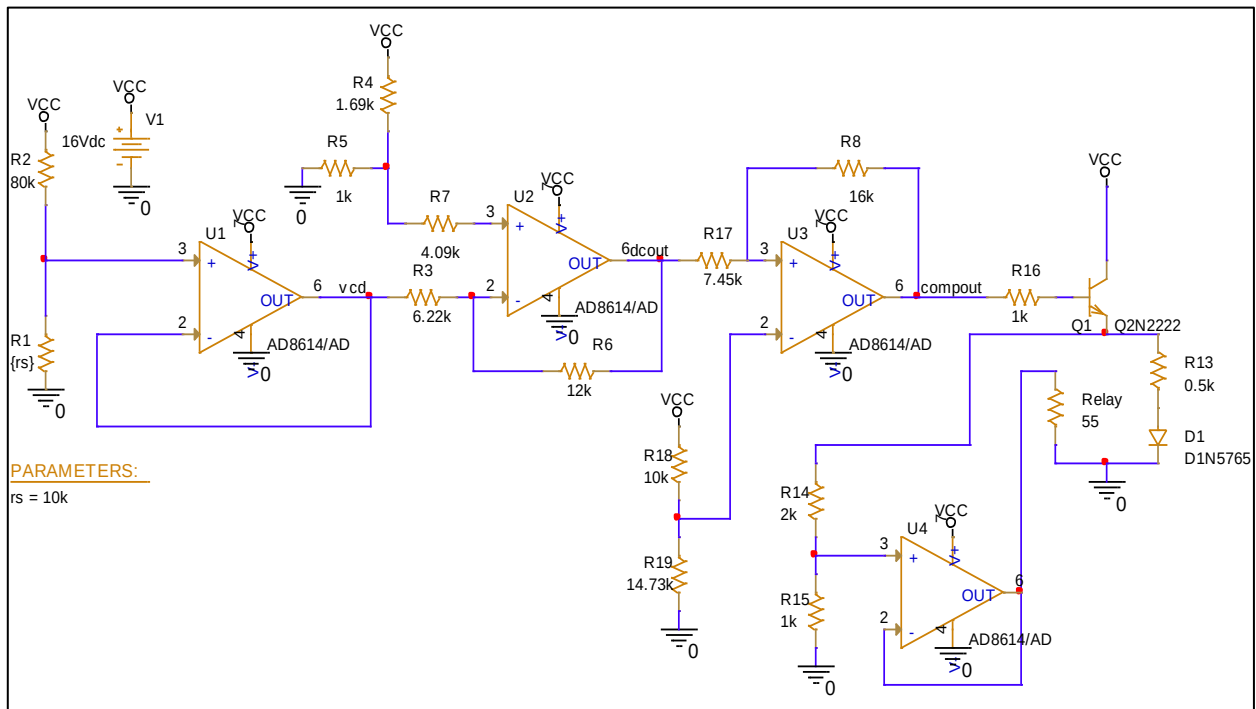


Figure 13 - Ideal Circuit Design

Trace Color	Trace Name	Y1	Y2	Y1 - Y2
	X Values	10.000K	43.793K	-33.793K
CURSOR 1,2	V(COMPOUT)	15.981	19.302m	15.961
	V(Relay:2)	5.0209	16.548m	5.0044
	V(DCOUT)	13.992	6.5023	7.4901

Figure 14 - Ideal Circuit Measurements

This diagram, however, represents the circuit in ideal conditions. The tolerances of the components will influence the circuit as shown in the next chapter.

STANDARDIZATION

For the standardization process, the rezistors used were designed to have the parameters of rezistors in E96 series, because of their low tolerance and high number of value variations.

- R_2 was set a tolerance of 1% and a value of $80.6k\Omega$
- R_3 was set a tolerance of 1% and a value of $6.19k\Omega$
- R_4 was set a tolerance of 1% and a value of $1.69k\Omega$
- R_5 was set a tolerance of 1% and a value of $1k\Omega$
- R_6 was set a tolerance of 1% and a value of $12.1k\Omega$
- R_7 was set a tolerance of 1% and a value of $4.12k\Omega$
- R_8 was set a tolerance of 1% and a value of $15.8k\Omega$
- R_{17} was set a tolerance of 1% and a value of $7.5k\Omega$
- R_{18} was set a tolerance of 1% and a value of $10k\Omega$
- R_{19} was set a tolerance of 1% and a value of $14.7k\Omega$
- R_8 was set a tolerance of 1% and a value of $15.8k\Omega$
- R_{20} was set a tolerance of 1% and a value of $2k\Omega$
- R_{21} was set a tolerance of 1% and a value of $1k\Omega$
- R_{16} was set a tolerance of 1% and a value of $5.36k\Omega$
- R_{13} was set a tolerance of 1% and a value of $0.5k\Omega$

The Op-Amps (AD8614) were chose because of their high-performance attributes, being both high-speed and high-precision. It offers low noise, which ensures better signal integrity. It is designed for a wide range of supply voltages, providing great flexibility, and features low power consumption. Also, they are single-supply and rail-to-rail Op-Amps, which offers us the possibility to make the switch of the circuit go from V_{cc} to 0 with high precision.

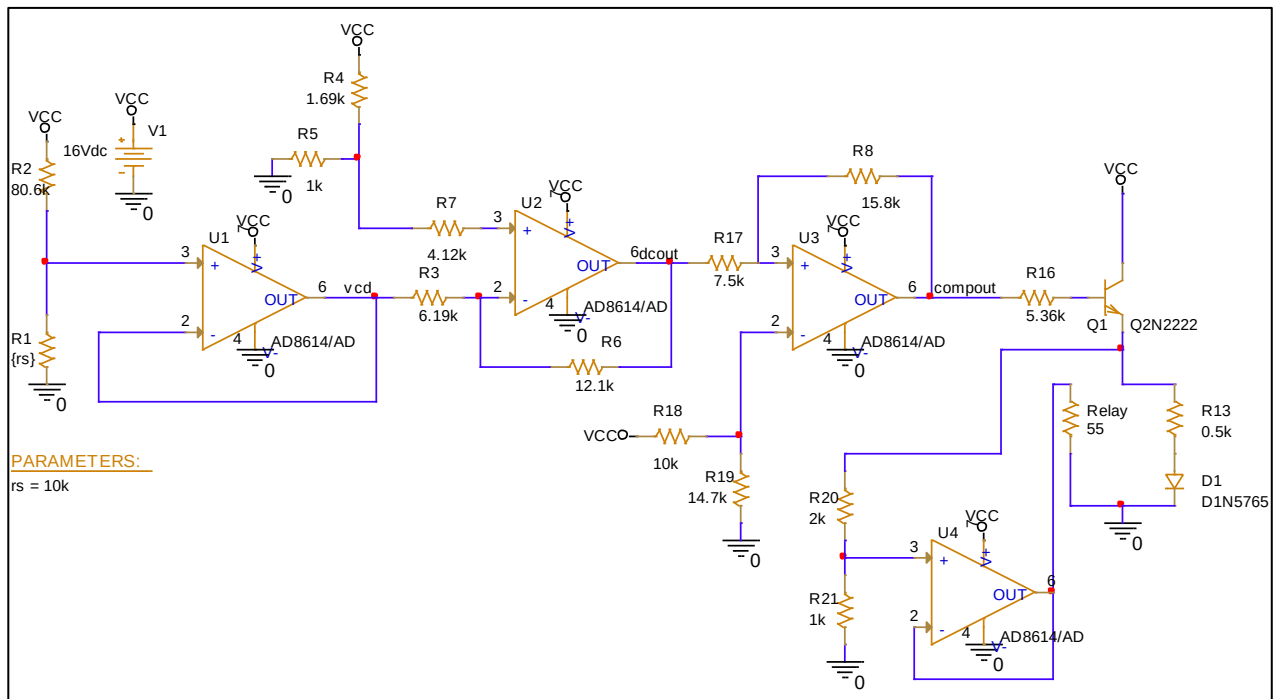


Figure 15 - Circuit scheme after standardization

Trace Color	Trace Name	Y1	Y2	Y1 - Y2
	X Values	10.049K	44.532K	-34.483K
CURSOR 1,2	V(Relay:2)	4.7794	16.548m	4.7629
	V(COMPOUT)	15.981	19.299m	15.962
	V(DCOUT)	14.107	6.4433	7.6633

Figure 16 - Circuit measurements after standardization

SIMULATIONS

- rs – global parameter to set the value of the sensor resistor to vary between $10k\Omega$ and $80k\Omega$.

1. DC Sweep

To analyze the output signal in the sensor resistance domain, I chose the DC Sweep analysis and utilized a parametric analysis to observe how the signal varies with different values of the parameter "rs".

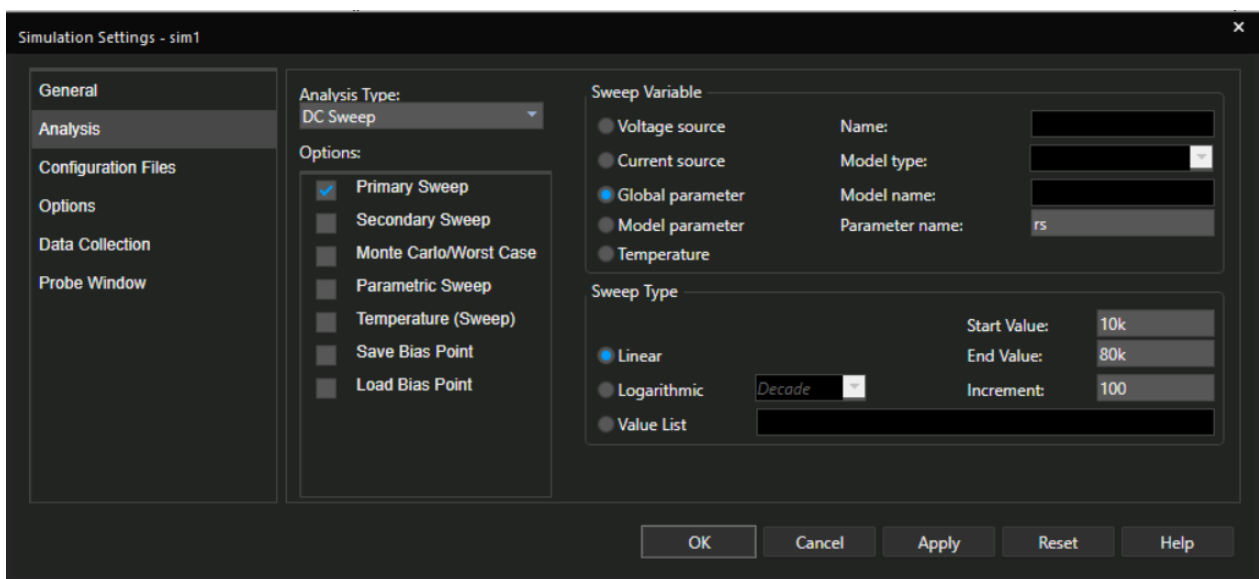


Figure 17 - DC Sweep Configurations

- I selected a step size of 100 due to its ability to provide a satisfactory level of accuracy while keeping the computation time relatively short.

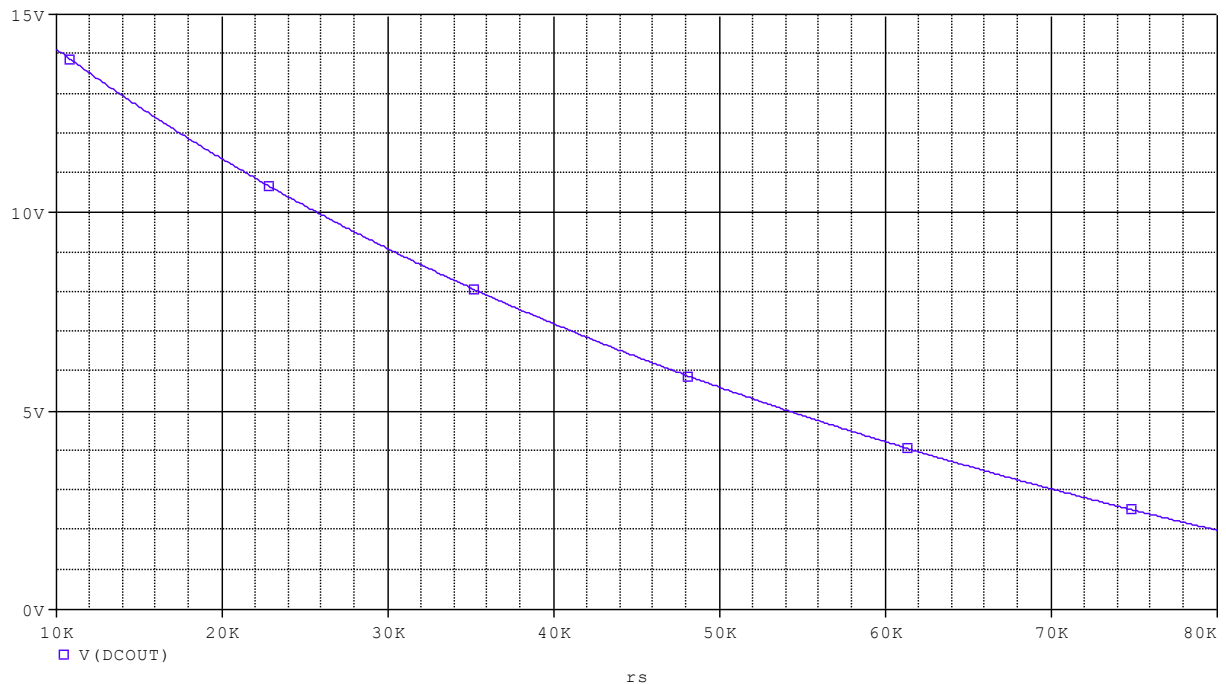


Figure 18 - Voltage variation at the converter's output

By looking at Figure 18 - Voltage variation at the converter's output we can observe the fact that the voltage variation is not perfectly linear because of the use of a voltage divider for the sensor and not another more precise circuit, for example an active bridge circuit, but for this application, a semi-linear graph is good enough.

If we plot the output of the comparator, we can see that the circuit switches off when the threshold voltage $V_{ThL} = 6.52V$ is reached (at the point where the sensor resistor reaches $r_s = 44.4k\Omega$).

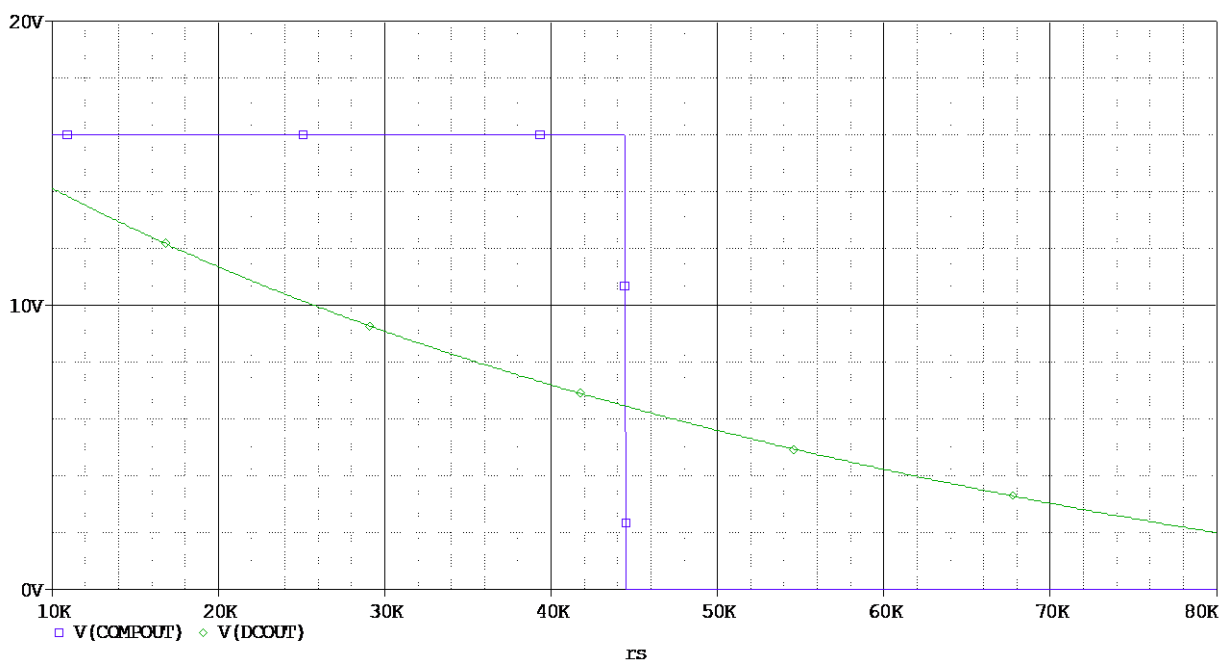


Figure 19 - Comparator output

If we edit the simulation profile and set Start Value to 80k Ω and End Value to 10k Ω , we can see the behavior of the circuit when the sensor resistance decreases (the concentration of carbon monoxide in the enclosure increases).

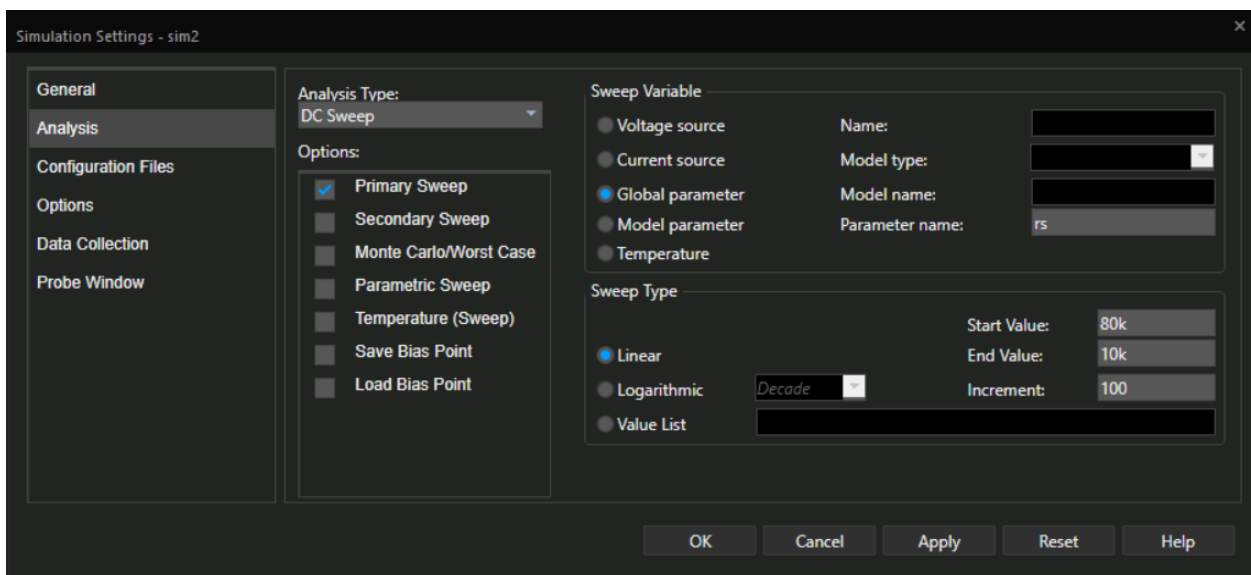


Figure 20 - New Simulation Profile

In the same manner as above, we are measuring the output of the comparator. If we plot the output of the comparator, we can see that the circuit switches on when the threshold voltage $V_{ThH} = 13.97V$ is reached (at the point where the sensor resistor reaches $rs=10.1k\Omega$).

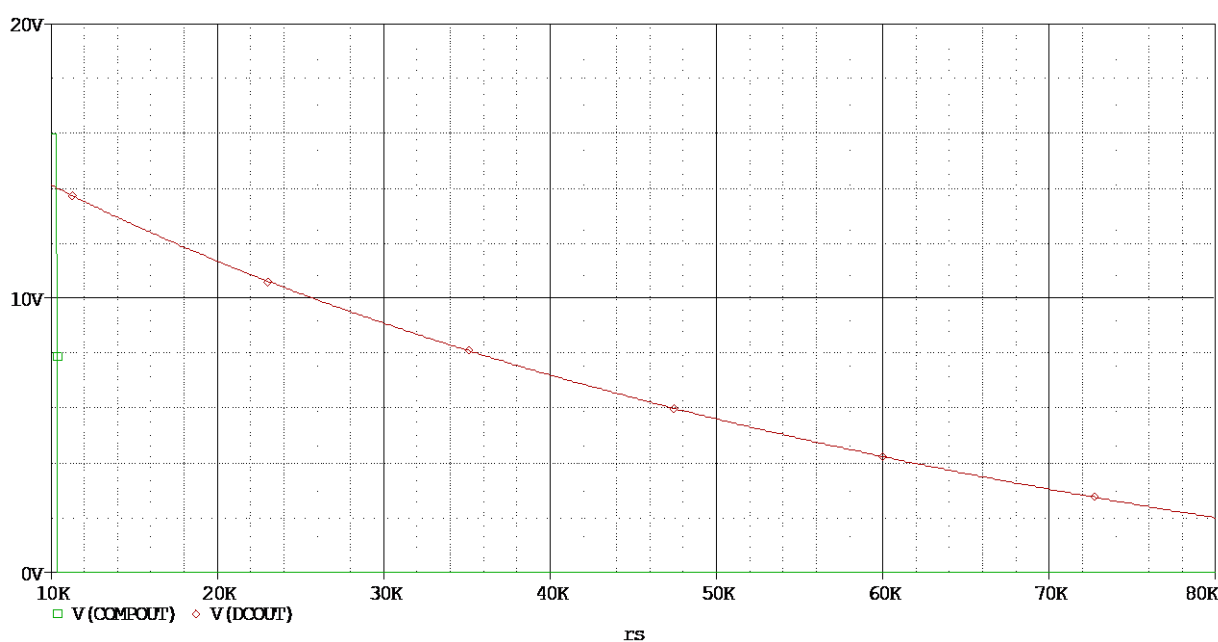


Figure 21 - New graph

By appending the graph from Figure 19 with the graph from Figure 21, we can see the whole hysteresis graph.

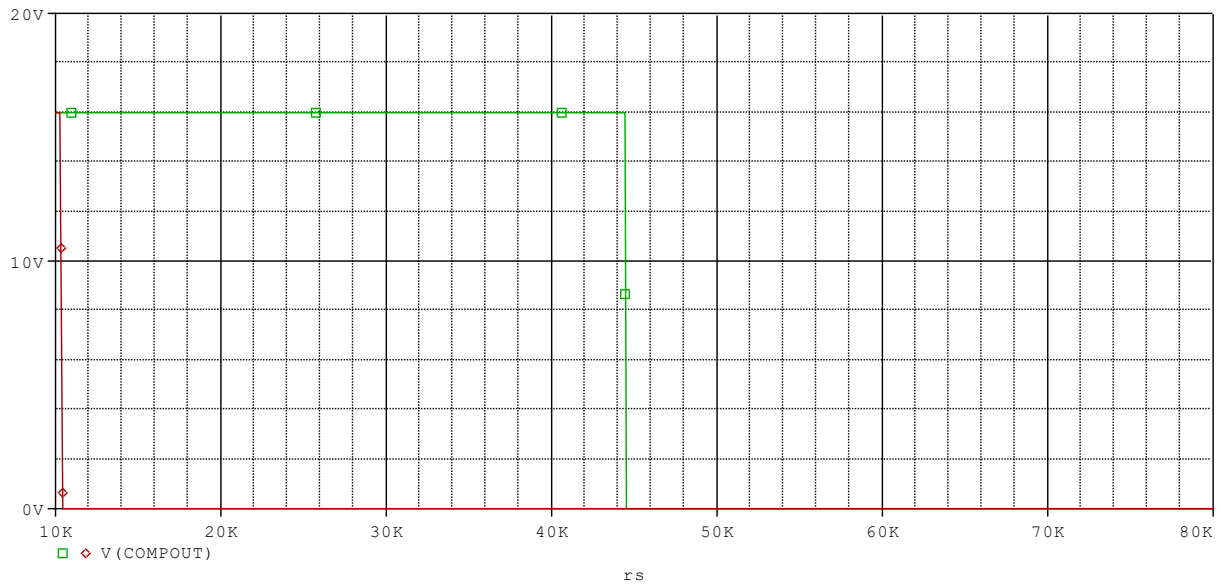


Figure 22 - Hysteresis

Because the relay is limited to 5V, the same behavior should be seen at the relay, but the highest voltage the circuit will have at that point will be 5V. We will verify this too because the relay will be the real output of the circuit.

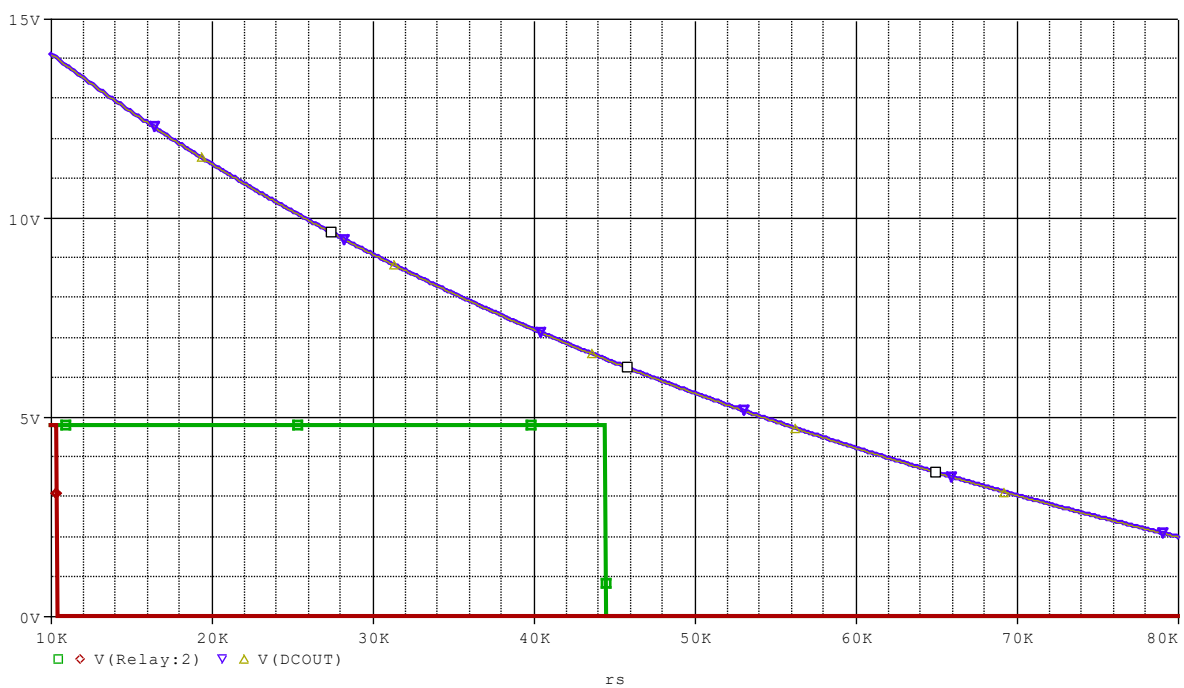


Figure 23 - Circuit output

Given the simulations done, we can see that the circuit works normally.

2. Temperature Sweep

Using this analysis, we can observe how the circuit behaves at different values for the temperature.

- Simulation Profile:

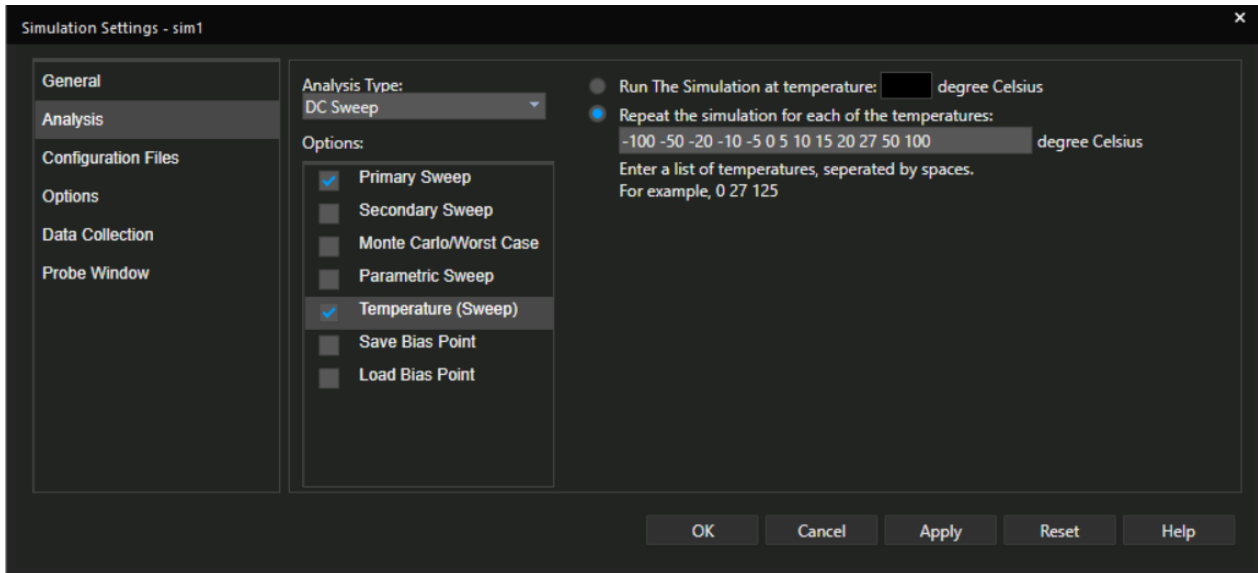


Figure 24 - Temperature Sweep Sim Profile

- Simulation Results:

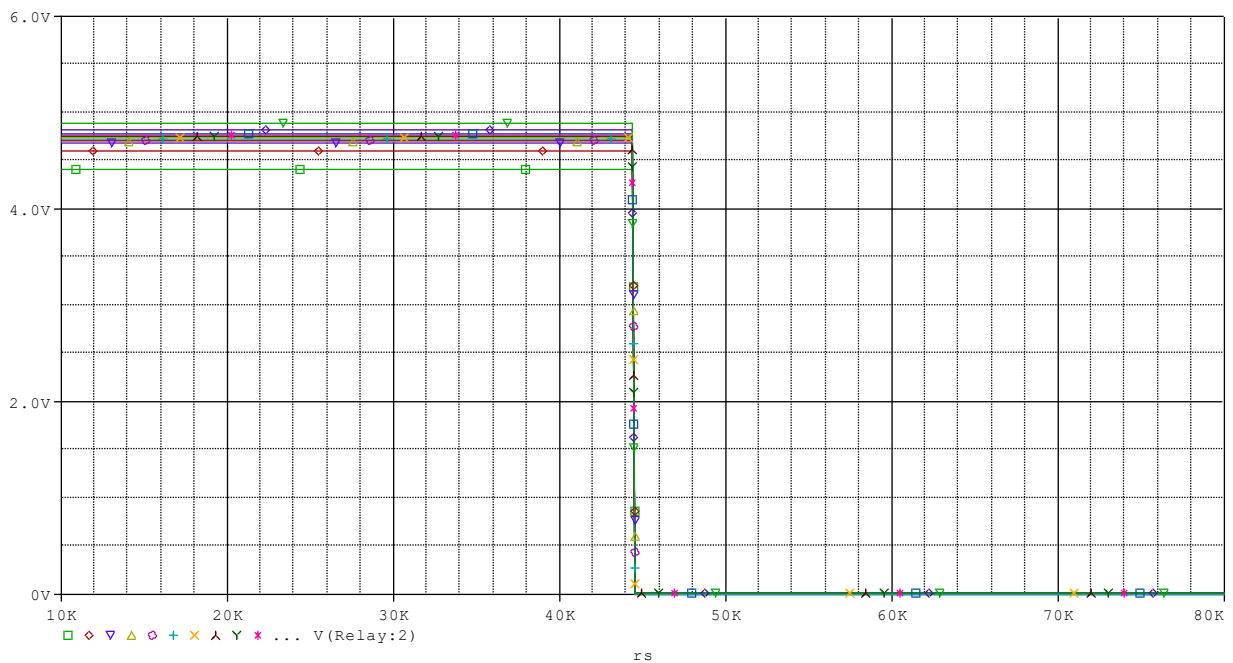


Figure 25 - Temperature Sweep Simulation

Evaluate	Measurement	1	2	3	4	5	6	7	8	9	10	11	12	13
☒	Min(V(Relay:2))	1.65463m	1.68805m	1.71009m	1.75724m	1.72279m	1.72873m	15.35137m	15.62348m	15.89546m	16.16730m	16.54765m	17.79549m	20.49824m
☒	Max(V(Relay:2))	4.41378	4.60152	4.68183	4.70504	4.71608	4.72679	4.73717	4.74724	4.75703	4.76654	4.77943	4.81861	4.89092

Figure 26 - Temperature Sweep Measurements

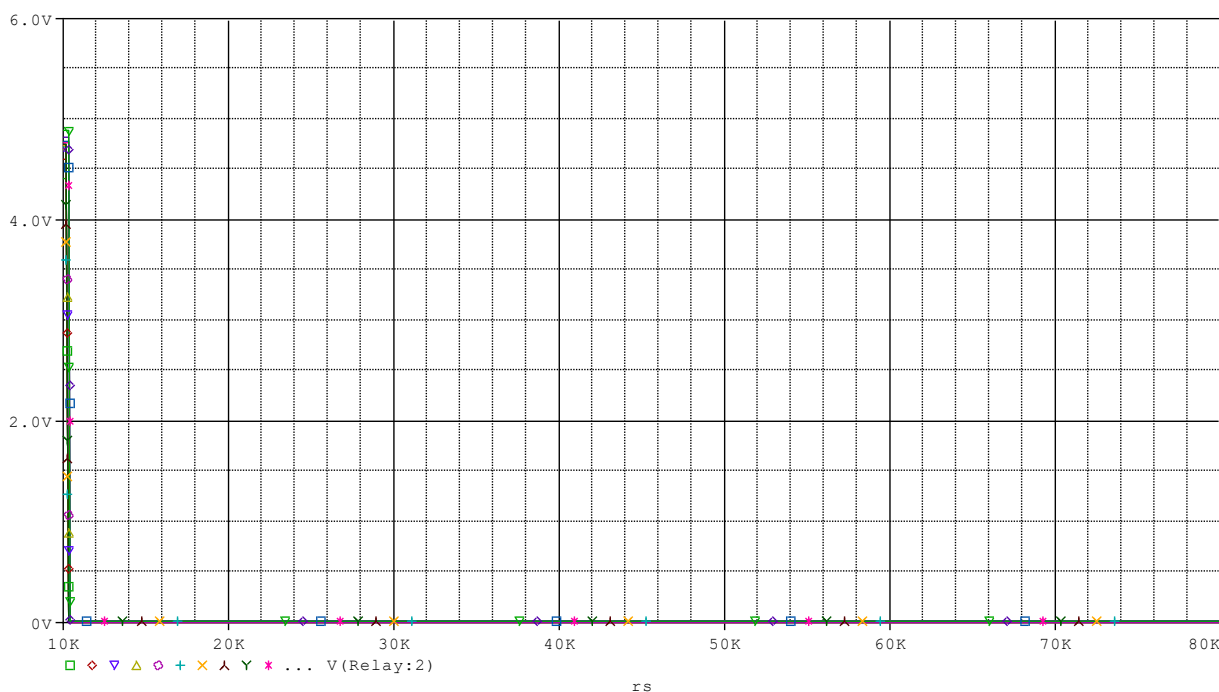


Figure 27 - Temperature Sweep Simulation

Evaluate	Measurement	1	2	3	4	5	6	7	8	9	10	11	12	13
☒	Min(V(Relay:2))	1.65463m	1.68806m	1.71008m	1.71817m	1.72280m	15.07912m	15.35137m	15.62348m	15.89546m	16.16730m	16.54765m	17.79549m	20.49824m
☒	Max(V(Relay:2))	4.41378	4.60152	4.68183	4.70504	4.71608	4.72679	4.73717	4.74724	4.75703	4.76654	4.77943	4.81861	4.89092

Figure 28 - Temperature Sweep Results

From Figure 25, Figure 26, Figure 27 and Figure 28, we can observe the fact that the maximum voltage the circuit reaches decreases with the increase of the temperature.

- Although the statement said above is true, the differences in voltage are very small in comparison with the differences in temperature, so we can say that the circuit is not influenced by temperature changes.

3. Monte Carlo analysis

Monte Carlo predicts the behavior of a circuit statistically when part values are varied within their tolerance range.

- Simulation profile:

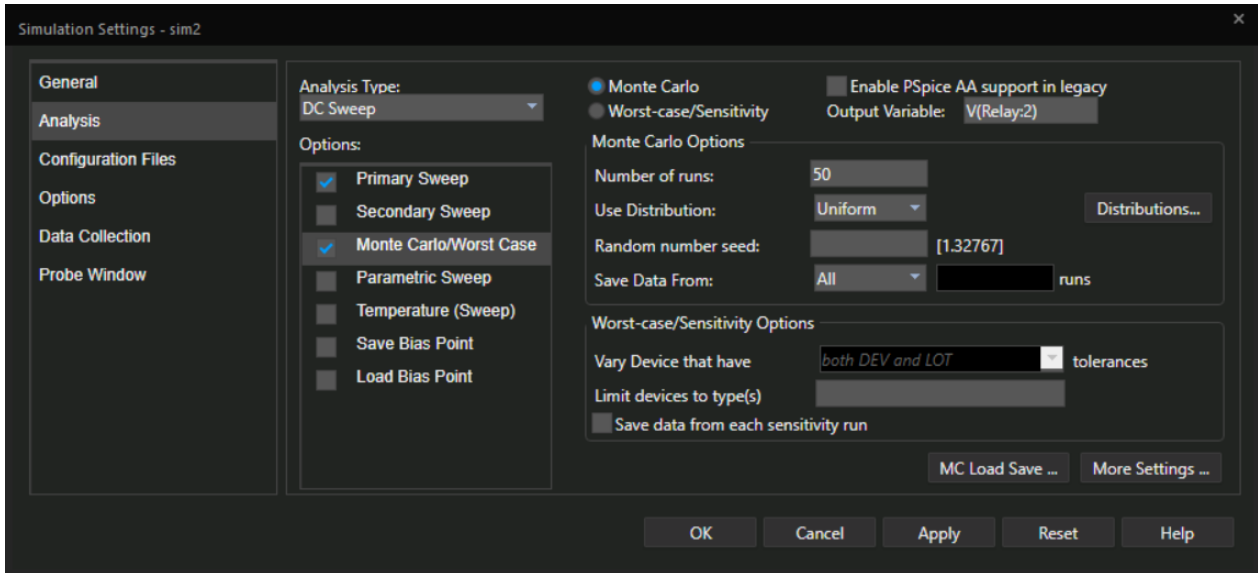


Figure 29 - Monte Carlo Profile

The Number of runs represents the total number of runs. The first run of the analysis is done at the nominal value of the components.

- Simulation results:

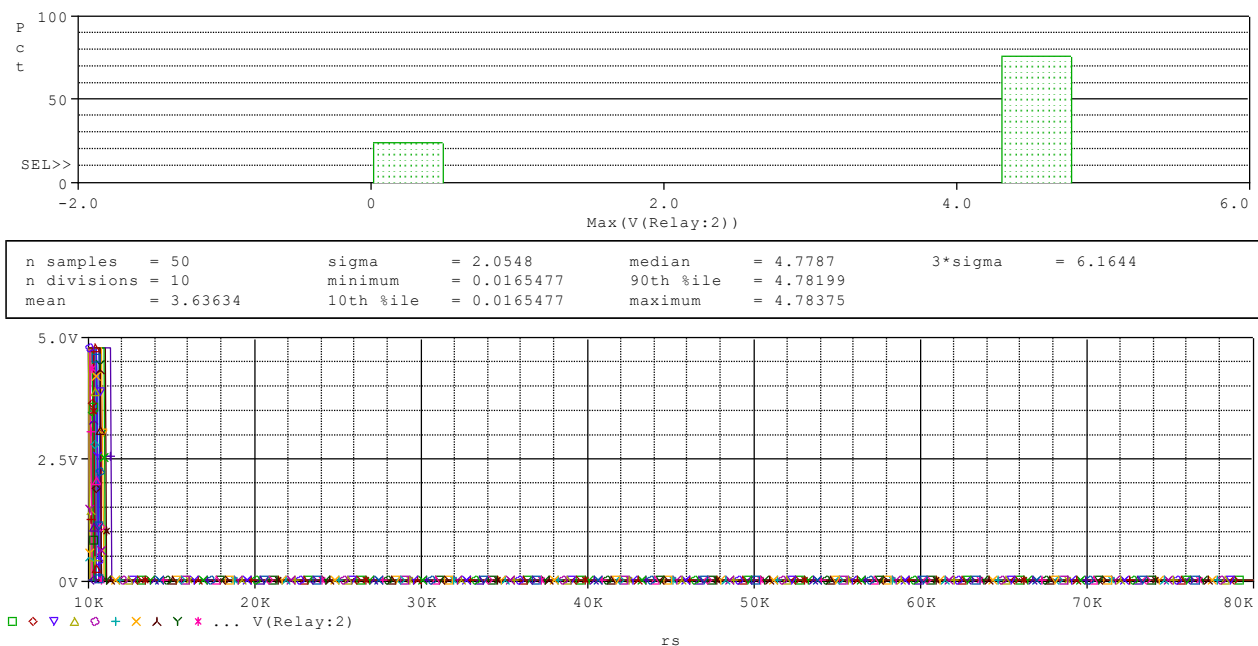


Figure 30 - 10k->80k Monte Carlo

- We can observe the tendency value of the signal to be 25% around 0.25V and 75% around 4.25V.

- This is not always true, because the values are chosen randomly for every run, and they may vary.

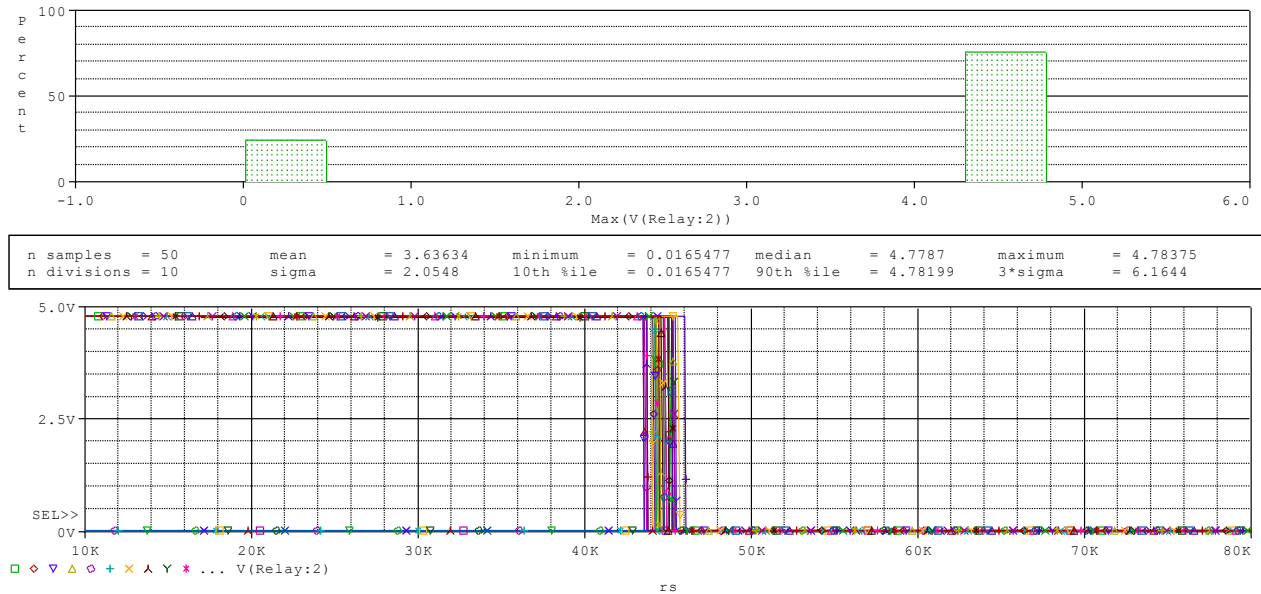


Figure 31 - 80k-10k Monte Carlo

- We can observe the tendency value of the signal to be 23% around 0.25V and 75% around 4.5V.
- This is not always true, because the values are chosen randomly for every run, and they may vary.

4. Worst Case Analysis

Sensitivity identifies which parameters are critical to the circuit operation. This analysis determines to what extent each component affects the circuit operation (single action or with other components). In addition, it changes all the values to simulate the worst case.

- Simulation profile:

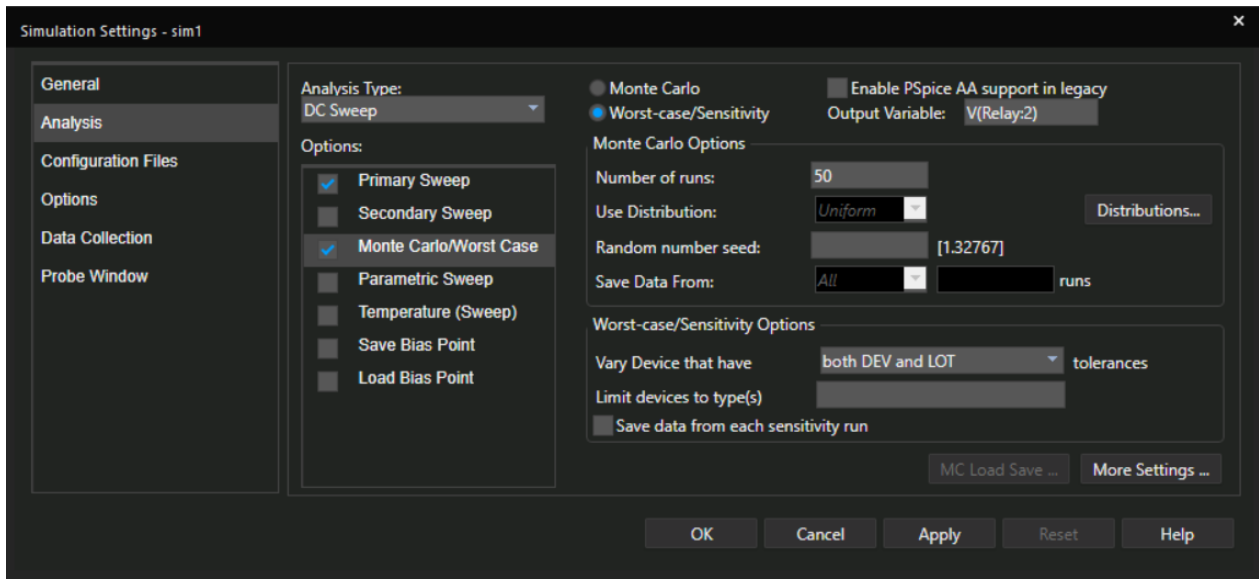


Figure 32 - Worst Case Profile

- Simulation results:

Sweeping r_s from 10k to 80k:

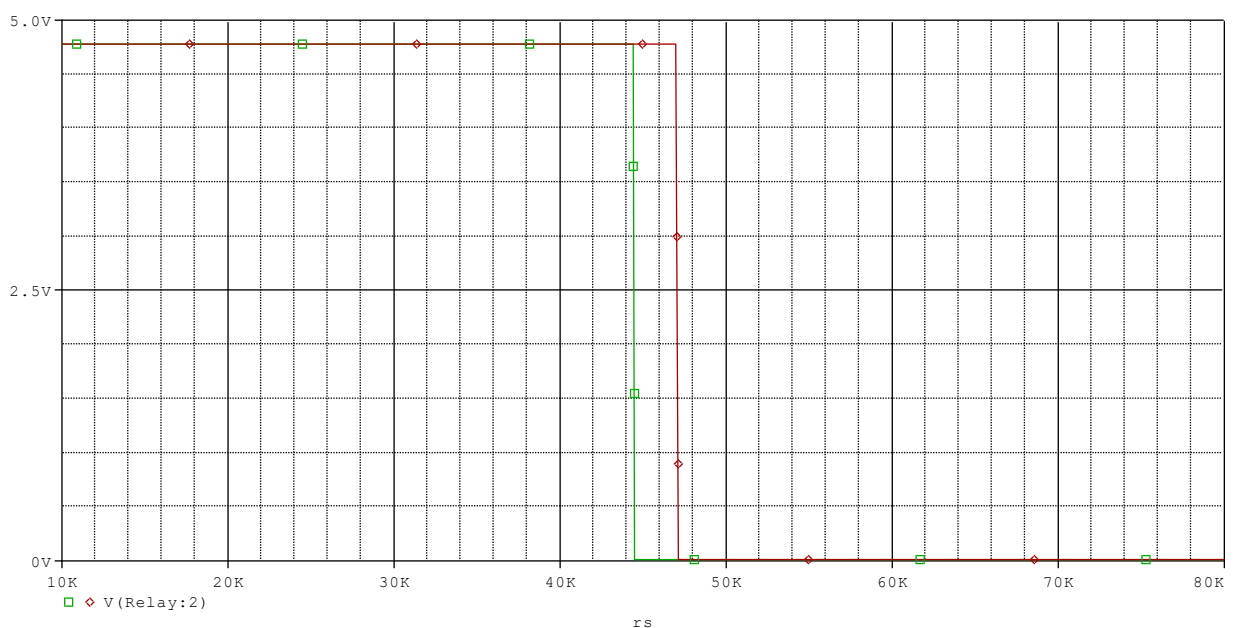


Figure 33 - 10k-80k Worst Case Results

- The green signal is the signal generated with nominal values
- The red signal is the worst case signal described by the output file as:

Device	MODEL	PARAMETER	NEW VALUE
R_R8	R_R8	R	1.01 (Increased)
R_R4	R_R4	R	.99 (Decreased)
R_R6	R_R6	R	1.01 (Increased)
R_R7	R_R7	R	.99 (Decreased)
R_R3	R_R3	R	.99 (Decreased)
R_R2	R_R2	R	1.01 (Increased)
R_R13	R_R13	R	1.01 (Increased)
R_R17	R_R17	R	1.01 (Increased)
R_R18	R_R18	R	1.01 (Increased)
R_R16	R_R16	R	.99 (Decreased)
R_R19	R_R19	R	.99 (Decreased)
R_R5	R_R5	R	1.01 (Increased)

Figure 34 10k-80k Resistor changes

Sweeping rs from 80k to 10k:

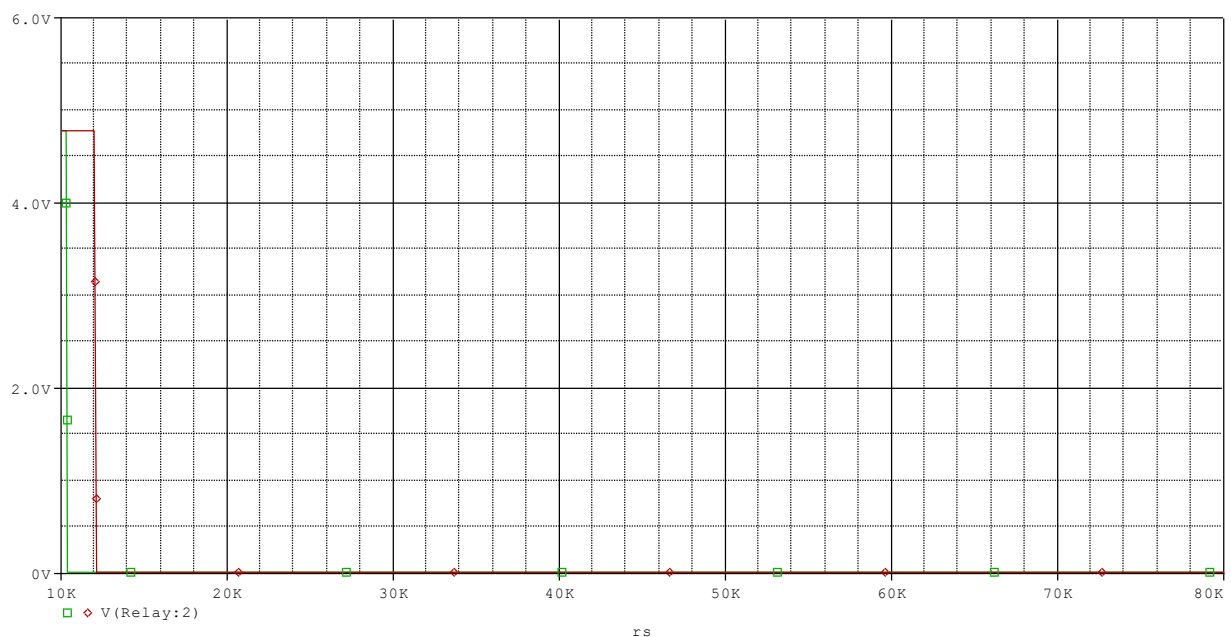


Figure 35 - 80k-10k Worst Case Sim

- The green signal is the signal generated with nominal values
- The red signal is the worst case signal described by the output file as:

Device	MODEL	PARAMETER	NEW VALUE
R_R8	R_R8	R	1 (Unchanged)
R_R4	R_R4	R	.99 (Decreased)
R_R6	R_R6	R	1.01 (Increased)
R_R7	R_R7	R	1 (Unchanged)
R_R3	R_R3	R	.99 (Decreased)
R_R2	R_R2	R	1.01 (Increased)
R_R13	R_R13	R	1.01 (Increased)
R_R17	R_R17	R	.99 (Decreased)
R_R18	R_R18	R	1 (Unchanged)
R_R16	R_R16	R	.99 (Decreased)
R_R19	R_R19	R	.99 (Decreased)
R_R5	R_R5	R	1.01 (Increased)

Figure 36 - 80k-10k Resistor changes

BILL OF MATERIALS

<i>Mfr. Part No.</i>	<i>Description</i>	<i>Quantity</i>	<i>Unit Price(EUR)</i>
<i>MRS25000C1002FC100</i>	10 k Ω	2	0,651
<i>CMF5080K600FHEB</i>	80.6 k Ω	1	0.725
<i>CCF606K19FKE36</i>	6.19 k Ω	1	1,61
<i>MBA02040C1691FC100</i>	1.69 k Ω	1	0,595
<i>MBB02070C7871FC100</i>	1 k Ω	2	0,586
<i>CCF6012K1FKE36</i>	12.1 k Ω	1	1,61
<i>CMF504K1200FHEB</i>	4.12 k Ω	1	0,381
<i>MF1/2DC1582F</i>	15.8 k Ω	1	0,664
<i>RNC50H7501FSB14</i>	7.5 k Ω	1	1,23
<i>LR1F5K36</i>	5.36 k Ω	1	0,14
<i>CCF5514K7FKE36</i>	14.7 k Ω	1	0,195
<i>RN60D2052FRE6</i>	0.5 k Ω	1	0,595
<i>MBB02070C7871FC100</i>	2 k Ω	1	0,586
<i>1N5765</i>	Red LED	1	(out of production)
<i>2N2222 PBFREE</i>	NPN Transistor	1	2,19
<i>AD8614</i>	Single 18V Op-Amp	4	4.86
<i>SRD-05VDC-SL-C</i>	5V Relay	1	0.98

CONCLUSION

The circuit designed for controlling the concentration of carbon monoxide (CO) in an enclosure successfully fulfills the specified requirements. The system effectively monitors CO levels using a resistive gas sensor and maintains the concentration within the predefined limits by activating a ventilation fan.

The simulations conducted, including DC sweep, temperature sweep, and Monte Carlo analysis, demonstrate the circuit's reliability and robustness under varying conditions. The circuit maintains functionality across different temperatures and component tolerances, indicating its suitability for practical applications.

Overall, the circuit design effectively addresses the task of controlling CO concentration in an enclosure, ensuring a safe environment by automatically managing ventilation based on real-time CO levels. The use of standard components and comprehensive simulation analysis further validate the circuit's performance and reliability.

This project serves as a practical implementation of electronic design principles, combining theoretical knowledge with practical application to solve a real-world problem. The successful design and simulation of this circuit highlight the importance of careful component selection, precise calculations, and thorough testing in electronic circuit design.

References

- [1 L. N. Ivanciu și E. Șipoș, Electronic Devices, Cluj-Napoca: UTPress, 2023.
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<http://bel.utcluj.ro/dce/didactic/ed/>
<http://bel.utcluj.ro/dce/didactic/fec/>
<https://eu.mouser.com/>
<https://www.arrow.com/en/products/1n5765/micropac-industries>
<https://www.digikey.ro/en/products/detail/analog-devices-inc/AD8614ARTZ-REEL7/621426>
<https://jansson.us/resistors.html>
https://components101.com/sites/default/files/component_datasheet/5V%20Relay%20Datash eet.pdf
<https://www.onsemi.com/pdf/datasheet/p2n2222a-d.pdf>
<https://ohmslawcalculator.com/voltage-divider-calculator>
<https://www.pspice.com/>
<https://www.alldatasheet.com/>